

δ -Aminolevulinic Acid Dehydratase Deficiency Porphyria

AT A GLANCE

- Extremely rare (fewer than 10 cases reported).
- Both early and late clinical onset have been described.
- Neurologic symptoms similar to those in other acute hepatic porphyrias can occur; no photosensitivity or cutaneous symptoms.

Markedly increased levels (usually 10- to 100-fold) of fecal **COPRO**, more than 90% of which is the **III isomer**, are present at all times in these patients. The feces may also contain increased amounts of hepta-, hexa-, and pentacarboxylic porphyrins. Urinary **COPRO III** is also elevated during attacks, as are levels of the precursors **ALA** and **PBG**. The latter may be normal between attacks.

HISTOPATHOLOGY

Skin lesions usually do not occur. When present, histopathologic features are similar to those in **PCT**.

DIFFERENTIAL DIAGNOSIS

In **HCP**, cutaneous signs are uncommon and, if they do occur, are most often seen in homozygous patients with disease onset in childhood. The skin manifestations resemble those seen in **PCT** and **VP**. Fecal predominance of **COPRO III** is more suggestive of **HCP** than of **VP**, in which **PROTO** and **COPRO** levels are usually increased.

Acute attacks are similar to those seen in **AIP**. The differential diagnosis rests on stool porphyrin determinations and measurement of **COPROGEN**

oxidase activity in fibroblasts or leukocytes. DNA analysis for mutations in the **COPROGEN oxidase gene** can also be performed.

TREATMENT

Treatment is the same as for **AIP** and **VP** and is described in Box 132-2.

δ-Aminolevulinic Acid Dehydratase Deficiency Porphyrria

ALA dehydratase deficiency porphyria is a very rare type of acute hepatic porphyria with an autosomal recessive inheritance pattern; fewer than 10 patients have been reported. Clinically, these patients present with symptoms similar to those of **AIP**. Attacks may be precipitated by stress and ethanol ingestion.

The disease results from a profound deficiency of **ALA dehydratase**, the second enzyme in heme biosynthesis, with enzyme activity less than 5% of normal. ALA dehydratase is a cytosolic enzyme that catalyzes the formation of **PBG** from two molecules of **ALA**.

The enzyme is encoded by the **ALA dehydratase gene** located on chromosome **9q34**. The mutations described to date are highly heterogeneous. Although initially thought to represent a homozygous defect, molecular studies have shown that at least one patient was a **compound heterozygote** for two independent mutant alleles.

Biochemical Findings

- **Urine:** Elevated levels of **ALA**, **COPRO**, and (to a lesser extent) **URO**.
- **Feces:** Variable findings — mildly elevated **COPRO** and **PROTO** in one patient, elevated **COPRO** in another, normal porphyrin profile in two others.
- **Erythrocytes:** **PROTO** level is elevated in all reported cases.

- **Plasma:** Increases in **ALA**, **COPRO**, and **PROTO** have been reported.

The porphyrin excretion pattern in these patients remains unexplained.

CONFIRMATION OF A SUSPECTED DIAGNOSIS OF PORPHYRIA

Diagnostic evaluation includes: - Comprehensive medical and family history - Clinical examination - Biochemical measurement of porphyrins and porphyrin precursors in **urine**, **feces**, and **blood** (see Fig. 132-21)

In specialized laboratories: - Specific enzyme activities can be measured in **erythrocytes**, **lymphocytes**, or **fibroblasts** - Molecular diagnosis via **DNA isolation** from peripheral blood, followed by **PCR amplification** and **DNA sequencing**

These techniques allow: - Precise identification of mutations - Functional analysis of genes - Accurate **genetic counseling** for patients and family members

Despite these advanced tools, diagnostic uncertainty may persist, particularly in cases of **dual porphyrias**, where overlapping biochemical features complicate interpretation.